• **VIBRATION—VIBRATION FELT AT HIGH SPEEDS—TRANSMISSION OUTPUT SHAFT BUSHING DAMAGED OR OUT OF POSITION—3.0L ENGINE 4X2 MODELS WITH MANUAL OR AUTOMATIC TRANSMISSION** • **TRANSMISSION—TRANSMISSION OUTPUT SHAFT BUSHING DAMAGED OR OUT OF POSITION—3.0L ENGINE 4X2 MODELS WITH MANUAL OR AUTOMATIC TRANSMISSION**	**Article No. 00-24-2**

FORD: 1998-2000 RANGER

This TSB is being republished in its entirety to revise the labor operation time and to update the Service Procedure and parts involved.

ISSUE

There may be a vibration at high speeds which may cause the transmission extension housing seal to leak or the transmission extension housing bushing to move out of position or become damaged on some vehicles. This may be caused by the calibration in the Powertrain Control Module (PCM).

ACTION

Replace the PCM, extension housing, and driveshaft and, if necessary, replace the transmission. The revised PCM assembly reduces the maximum speed limit to 150 km/h (94 mph) for Flex Fuel Vehicles (FFVs) and 145 km/h (90 mph) for gasoline vehicles in U.S. applications which will reduce the possibility of vibrations. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

Refer to the appropriate Ranger Workshop Manual Sections for normal diagnostics and removal and installation procedures for the automatic or manual transmission.

1. Perform quick tests and record all Diagnostic Trouble Codes (DTCs).
2. Repair all DTCs per the Workshop Manual.
3. Inspect the transmission for a fluid leak at the extension housing and proceed as follows:
   a. Inspect driveshaft alignment marks for alignment and remove driveshaft. If alignment marks are not aligned, record variation on the back of the repair order for future reference.
   b. AUTOMATIC TRANSMISSION ONLY: Inspect extension housing bushing for damage or out-of-location within the bore for one of the following two scenarios:

SCENARIO #1:

- If extension housing bushing IS in the proper location, remove the transmission oil pan, inspect fluid for contamination and determine if fluid is burnt.

- If HEAVY contamination is present or fluid is burnt, replace the Transmission (F87Z-7000-JARM for 1998 models or XL5Z-7000-AARM for 1999-2000 models). Flush the transmission cooler line system forward and backward (lines only). Replace Radiator (XL2Z-8005-LA).

- If MINIMAL contamination is present, replace the Extension Housing (F77Z-7A039-CA) and Gasket (D4ZZ-7086-B). Clean the transmission oil pan and install a new transmission fluid pan gasket. Reinstall transmission oil pan and flush transmission cooling system forward and backward (lines and cooler).

SCENARIO #2:

- If extension housing bushing IS NOT in the proper location within the bore, replace the Transmission (F87Z-7000-JARM for 1998 models or XL5Z-7000-AARM for 1999-2000 models).

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- If HEAVY contamination is present, flush the transmission cooler line system forward and backward (lines only) and replace Radiator (XL2Z-8005-LA).

- If MINIMAL contamination is present, flush transmission cooling system forward and backward (lines and cooler).

c. MANUAL TRANSMISSION ONLY: Inspect extension housing bushing for damage or out-of-location within the bore for the following two scenarios:

SCENARIO #1:

- If extension housing bushing IS in the proper location, drain the transmission oil, inspect fluid for contamination, and determine if fluid is burnt.

- If HEAVY contamination is present or fluid is burnt, replace the Transmission with a 2001 model year transmission (1L54-7003-CA).

NOTE
WHEN REPLACING THE TRANSMISSION WITH A 2001 MODEL YEAR TRANSMISSION YOU WILL HAVE TO REPLACE THE EXISTING EXTENSION HOUSING WITH (F2TZ-7A039-A), AND YOU WILL ALSO HAVE TO REMOVE THE SENSOR ROTOR FROM THE NEW TRANSMISSION WHICH IS LOCATED ON THE MAIN SHAFT.

- If MINIMAL contamination is present, drain the transmission fluid, replace the Extension Housing (F2TZ-7A039-A), use Ford-approved RTV Gasket Sealant (F7TZ-19554-AA, or equivalent meeting Ford specification ESB-M4G92-A) and refill with MERCON ®V Automatic Transmission Fluid (XT-5-QM).

SCENARIO #2:

If extension housing bushing IS NOT in the proper location within the bore, replace the Transmission with a 2001 model year transmission (1L54-7003-CA).

NOTE
WHEN REPLACING THE TRANSMISSION WITH A 2001 MODEL YEAR TRANSMISSION YOU WILL HAVE TO REPLACE THE EXISTING EXTENSION HOUSING WITH (F2TZ-7A039-A). AND YOU WILL ALSO HAVE TO REMOVE THE SENSOR ROTOR FROM THE NEW TRANSMISSION WHICH IS LOCATED ON THE MAIN SHAFT.

d. AUTOMATIC AND MANUAL TRANSMISSIONS CONTINUED:

- Install a new driveshaft and make sure that the driveshaft and axle alignment marks are aligned. Refer to Parts Block for proper driveshaft usage.

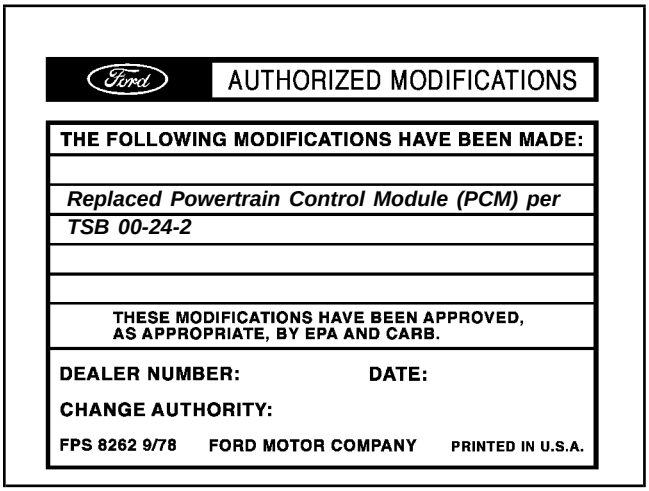
- Fill transmission fluid to the proper level using MERCON®V Automatic Transmission Fluid (XT-5-QM).

- Replace the PCM. For proper removal and installation procedures, refer to your appropriate Ranger Workshop Manual. Refer to chart that follows for proper PCM usage.

4. Road test vehicle to verify repair.

PCM REPLACEMENT INFORMATION				
Application	**Old Part Number (-12A650-)**	**Old Tear Tag**	**New Part Number (-12A650-)**	**New Tear Tag**
FFV				
1999 A/T 4X2 Ranger FFV/CAL	XL5F-KA, KC XU2F-HA	KDM0, 2 / NCH0	XU2Z-BCA	WTV0
1999-1/4 to 2000 A/T 4X2 Ranger FFV/CAL	XL5F-KD, KE, KF, KG	KDM3, 4, 5, 6	YU7Z-MA	KZG0
1999 A/T 4X2 Ranger FFV/FED/ALT/CAN	XL5F-HA, HC XU2F-FA	HCL0, 2 / MJH0	XU2Z-BBA	HEH0
1999-1/4 to 2000 A/T 4X2 Ranger FFV/FED/ALT/CAN	XL5F-HD, HE, HF, HG	HCL3, 4, 5, 6	YU7Z-UA	QGF0
1999/2000 M/T 4X2 Ranger FFV/FED/ALT/CAN	XL5F-MA, C, D, E, F, G	MXA0, 2, 3, 4, 5, 6	YL5Z-ZA	FHZ0
1999/2000 M/T 4X2 Ranger FFV/CAL	XL5F-YA, C, D, E, F, G	YBJ0, 2, 3, 4, 5, 6	YL5Z-ACA	EZU0
GAS				
1998 M/T 4X2 Ranger 49S/ALT/CAN	F87F-XA, B	EFA0, 1	F8PZ-EPA	WSN0
1998 M/T 4X2 Ranger CAL	F87F-AAA, B	HND0, 1	F8PZ-ETA	YPK0
1998 A/T 4X2 Ranger 49S/ALT/CAN	F87F-BNA	BPE0	F8PZ-EKA	KSK0
1998 A/T 4X2 Ranger CAL	F87F-BSA	EMR0	F8PZ-ENA	TXW0
1999 A/T 4X2 Ranger CAN	XL5F-FA, B	JAA0, 1	XU2Z-AZA	YVR0
1999/2000 M/T 4X2 Ranger CAN	XL5F-AGA, B	MBJ0, 1	YL5Z-AFA	NXV0
1999-1/4 to 2000 A/T 4X2 Ranger CAN	XL5F-ATA, B	LLF0, 1	YU7Z-RA	FLU0

Obtain an Authorized Modifications Decal (FPS 8262 - obtainable through DOES II, 25/pkg) and list the date, dealer number, and summary of modifications performed. Select a prominent place adjacent to the Vehicle Emission Control Information Decal suitable for installing the Authorized Modifications Decal. Clean the area, install the decal, and cover it with a clear plastic decal shield.



Article No. 00-24-2 Cont'd.

PART NUMBER	PART NAME
F87Z-4602-AA	Driveshaft (112″ Wheelbase - Automatic Transmission)
F87Z-4602-EA	Driveshaft (118″ Wheelbase - Automatic Transmission)
F87Z-4602-PA	Driveshaft (126″ Wheelbase - Automatic Transmission)
XT-5-QM	MERCON®V Automatic Transmission Fluid
F5TZ-7A191-A	Gasket - Transmission Fluid Pan
F77Z-7A039-CA	Extension Housing (Automatic Transmission)
D4ZZ-7086-B	Gasket - Extension Housing (Automatic Transmission)
F87Z-7000-JARM	Transmission Assembly - Automatic (1998 4X2 With 3.0L Engine)
XL5Z-7000-AARM	Transmission Assembly - Automatic (1999/2000 4X2 With 3.0L Engine)
XL2Z-8005-LA	Radiator
F87Z-4602-CA	Driveshaft (112″ Wheelbase - Manual Transmission)
F87Z-4602-GA	Driveshaft (118″ Wheelbase - Manual Transmission)
F87Z-4602-NA	Driveshaft (126″ Wheelbase - Manual Transmission)
F2TZ-7A039-A	Extension Housing (Manual Transmission)
F7TZ-19554-AA	RTV Gasket Sealant
1L54-7003-CA	Transmission Assembly (5-Speed Manual)

OTHER APPLICABLE ARTICLES: NONE
SUPERSEDES: 00-15-1
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage And Emissions Warranty Coverage

OPERATION	DESCRIPTION	TIME
002402A	(A/T) Inspect Trans For Leaks, Remove & Install Trans Pan-Inspect For Contamination, Inspect/Record Driveshaft Alignment Marks, Replace Extension Housing, Flush Trans Cooler, Replace PCM And Driveshaft (Includes Time To Road Test)	2.9 Hrs.
002402B	(A/T) Inspect Trans For Leaks, Remove & Install Pan-Inspect For Contamination, Inspect/Record Driveshaft Marks, Remove/Inspect Trans Extension Housing, Replace Trans, Flush Trans Cooler, Replace PCM & Driveshaft (Includes Road Test)	4.5 Hrs.
002402C	Extra Time To Replace Radiator If Required (To Be Used Only With Operation B)	0.5 Hr.
002402D	(M/T) Inspect Trans For Leaks, Inspect/Record Driveshaft Alignment Marks, Replace Extension Housing And Inspect For Contamination, Drain Trans Fluid, Replace PCM And Driveshaft (Includes Road Test)	2.1 Hrs.
002402E	(M/T) Inspect Trans For Leaks, Inspect/Record Driveshaft Alignment Marks, Remove Extension Housing And Inspect For Contamination, Replace Trans, Replace PCM And Driveshaft (Includes Road Test)	3.9 Hrs.
002402F	Extra Time To Replace Extension Housing And Remove Sensor Ring On Manual Trans (To Be Used Only With Operation E)	0.4 Hr.

DEALER CODING

BASIC PART NO.	CONDITION CODE
12A650	42

OASIS CODES: 504000, 505000, 510000, 703000, 703300